



A game-theoretic model of misinformation spread on social networks

Chin-Chia Hsu^{ID,*,1}, Amir Ajorlou^{ID,2}, Ali Jadbabaie^{ID,2}

Institute for Data, Systems, and Society, Massachusetts Institute of Technology (MIT), Cambridge, MA, 02139, USA

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ABSTRACT

In this paper, we develop a game-theoretic model of sharing decisions among users of a Twitter-like social network. Agents have a subjective prior on an unobservable real-valued state, representing their beliefs on a topic that is subject to a binary action. Agents cast their binary action, matching the sign of the state. Before the action stage, a small fraction of agents receive a piece of news which impacts their beliefs. Those who receive the news update their belief and make a decision as to whether to share the news with their followers in order to influence their beliefs, and in turn their actions, given a fixed cost for sharing. We characterize the underlying news spread as an endogenous Susceptible-Infected (SI) epidemic process and derive agents' sharing decisions as well as the size of the sharing cascade at the equilibrium of the game. We show that lower credibility news can result in a larger cascade than fully credible news provided that the network connectivity surpasses a *connectivity limit*. We further delineate the relationship between cascade size, network connectivity, and news credibility in terms of polarization and diversity in prior beliefs: We demonstrate that increased polarization reduces the connectivity limit whereas larger in-party diversity has a non-monotone effect on the connectivity limit, which depends on both the levels of polarization and in-party diversity. Our results provide a theoretical foundation for recent empirical observations demonstrating faster and wider spread of low-credibility and false information on social networks.

1. Introduction

Social networks have become a major source of news and information for a large segment of the population across the political spectrum. In particular, social media giants such as Facebook and Twitter have amassed a large number of users who primarily consume their news via these platforms. Furthermore, publishers increasingly use social media to inform the public of unknown facts or the latest news (Shearer and Matsa, 2018). Politicians also launch their political propaganda on social media more often than via conventional channels to promulgate their ideology and mobilize voters (Williams and Gulati, 2013). A myriad of various information is circulating on online social networks.

Despite many of their benefits in connecting people, social media have also become a conduit for spread of inaccurate news and misinformation. Many popular and widespread posts or memes on social media continue to be inaccurate, misleading, or false (Gill, 2018).

* Corresponding author.

E-mail addresses: chinchia.k.hsu@gmail.com (C.-C. Hsu), ajorlou@mit.edu (A. Ajorlou), jadbabai@mit.edu (A. Jadbabaie).

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2017; Silverman et al., 2017; Silverman and Pham, 2018). This is due to the fact that any user on social media can readily publicize digital content and share others' posts, often without verification, fact-checking, or significant third party filtering.

A flurry of recent empirical findings (Takayasu et al., 2015; Vosoughi et al., 2018) suggest that false news and misinformation are more likely to disseminate on social networks, spreading faster, deeper, and more widely than real information. A recent study (Juul and Ugander, 2021) shows that difference between false-news and real-news dissemination may be substantially explained by the difference in peer-to-peer transmission rate of news. Even though recent literature across multiple disciplines has outlined the impact of prevalent misinformation on public opinions, especially on elections (Bessi and Ferrara 2016; Allcott and Gentzkow 2017), little progress has been made on why and how inaccurate information can trigger a large sharing cascade among users on social media.

In order to provide a theoretical framework for such empirical findings, in this paper we consider a simple, game-theoretic setting that is focused on news-sharing decisions of individuals on social networks. Using the proposed game-theoretic model, we aim to address several fundamental questions regarding news dissemination on social networks: (i) Under what conditions does a piece of news that initially goes to only an infinitesimal subset of the population spread to a significant fraction of the population? (ii) When does low credibility news spread wider than highly credible news? (iii) What is the association between population characteristics, news credibility and size of news spread?

Instead of making a dichotomy between true and false news, this paper concerns news that is imperfect information about an underlying state of the world. This corresponds to the scenarios in which the state has not been uncovered nor identified at the time of publishing articles. The extent to which agents take into account the news varies with the news agencies' investment in estimating the state, or relatedly, their credibility in news reporting. We focus on news with credibility that is known to the population, corresponding to the old news media who possess long history and well-established reputation in the news market.

At the core of our study lies the motivation of individuals behind news sharing. Prior theoretical and empirical research identifies four key factors: (1) the maintenance of social relationships, (2) the pursuit of psychological benefits (e.g., entertainment value), (3) the acquisition of peer awards (e.g., likes, recognition, and reputation), and (4) intrinsic or altruistic motives, including informing or persuading others (Lee and Ma, 2012; Ma and Chan, 2014; Kümpel et al., 2015; Oh and Syn, 2015; Wang and Rzeszutarski, 2023). This paper examines how altruistic motives influence news sharing, specifically when individuals aim to *persuade* their peers towards actions they believe are beneficial. Consider a topic of common interest, such as mask-wearing, where an individual cares about their peers' actions. The persuasion motive arises since an individual's belief about the topic may differ from her peers' beliefs and the optimal actions can be subjectively dissimilar. When receiving some new information, the individual can influence her peers' beliefs and their subsequent actions by strategically sharing the information and altering their information sets. Likewise, one may choose to withhold the information if it otherwise persuades her peers to take undesired actions from her view.

Conversely, individuals may exhibit hesitancy in sharing or posting content on social media. A Facebook post or a Tweet can, at times, ignite intense discussions or even toxic debates. Research indicates that users on social media are increasingly aware of the potential negative repercussions stemming from their shared content (Newman et al., 2018, 2019). Surveys of U.S. adults (Duggan, 2017; Vogels, 2021) reveal that social media platforms are the primary venues for online harassment, and that the prevalence of such harassment is increasing. These experiences can lead to feelings of regret and negatively impact mental health, prompting users to hesitate, self-censor, or abstain from sharing content (Wang et al., 2011; Sheldon et al., 2019). However, despite these concerns, content sharing and retweeting remain prevalent on social platforms. This suggests that many users are still willing to post and share when they perceive sufficient value in doing so.³

Balancing the persuasive motive with concerns about potential negative consequences, we examine the decision process of news sharing among a continuum of agents connected on a social network. The agents have heterogeneous subjective prior beliefs on an unobservable real-valued state. Each agent will ultimately take a binary action, such as to wear or not to wear a mask, that aligns the sign of the state based on their individual beliefs. Nature generates a single piece of real-valued news about the state, along with its credibility (representing the sensitivity of agents' beliefs to the news). Initially, a small, randomly selected subset of the population is informed of this news and its credibility.

The network that depicts social interactions within the population is exogenously constructed using a simple random graph model with a Poisson distribution for the number of one's followees.⁴ Once the social network is realized, we assume that each agent only knows her followers and followees but has no knowledge about interactions among others. This modeling approach for network games is similar to the theoretical frameworks proposed in Galeotti and Goyal (2009); Fainmesser and Galeotti (2016, 2020), enabling us to perform tractable analysis of otherwise intractable diffusion processes on social networks (Newman, 2003; Jackson, 2008). This notion that agents' knowledge of the network are limited to their neighbors is also empirically corroborated by Breza et al. (2018).

Upon observing the news and its credibility, an agent updates her belief, fusing her prior belief with the news. Subsequently, she chooses whether to share the news with all of her followers (e.g., retweeting the news), deliberating how effectively the news can persuade her followers to take the binary action in line with her belief. We assume that news sharing incurs a cost, which captures an individual's apprehension regarding potential unpleasant public discussions that may follow after her social media post. Due to this cost associated with sharing, agents' sharing decisions feature strategic substitutability. Specifically, if an agent speculates that her

³ Another concern that can deter individuals from publicly expressing their opinions is the belief that those opinions are in the minority, coupled with a fear of social isolation. This is the core concept of the "spiral of silence" theory (Noelle-Neumann, 1974), which posits that this silence reinforces perceived dominance of the majority opinion, further suppressing the opinion of minority in a spiraling effect. Several empirical studies have examined this theory in the context of social media platforms (Hampton et al., 2014; Stoycheff, 2016).

⁴ We abstract the network formation using random network modeling, unlike the set of models (Sethi and Yildiz, 2016; Hsu et al., 2020b) that endogenized the social network as a result of individual strategic behavior of following or connecting to others in order to garner relevant information.

followers are likely to hear the news through other agents, she may choose not to share to avoid the cost. As a result, each agent has to strategically evaluate the impact of *other agents' news-sharing decisions* on her followers' information sets while making her sharing decision.

As our first result, we characterize the news-sharing strategies in the game. The sharing rule is a function of an agent's belief mean about the state and the effective persuasiveness power of the news: Agents with more extreme beliefs have stronger incentives (or stronger disincentives, depending on whether the news is in alignment with their beliefs) to share news. We establish the system of equations that govern the interdependency between individual sharing strategies and the scale of news spread that endogenously emerges. Using the equations we quantify the cascade size at the equilibrium and show the necessary and sufficient conditions for occurrence of a news cascade. Given news credibility, we show that more extreme news increases the spread size and that news spreads wider on more connected networks.

As our second contribution, we identify the necessary and sufficient condition under which news with lower credibility can trigger a larger cascade on social networks. We further discover that when news is quite neutral to both actions, lower credibility can make news diffusion larger than full credibility. The beliefs of informed agents about the state are concentrated around the small news value when observing high level of credibility. Consequently they find insufficient benefit of making their followers' actions aligned with the sign of the state, having weak incentives to persuade their followers. By contrast, though lower credibility undermines the persuasiveness of news, the extremist agents still hold relatively extreme beliefs about the state and prefer to share, leading to a larger cascade than credible news.

The condition also suggests that fully credible news no longer generates a larger cascade of news sharing than lower credibility news when connections in a network are sufficiently dense. Specifically, there is a limit on network connectivity (the average number of followers) beyond which lower credibility news spreads wider than fully credible news. This result is rooted at the strategic substitutability of sharing decisions: The ease of diffusion caused by greater connectivity suggests that informed agents share the news only if they hold more extreme beliefs and, in turn, stronger sharing incentives. For news with lower credibility, since it does not move the beliefs of extremists towards the news value as much as fully credible news, more fraction of informed agents hold extreme beliefs and participate in sharing, resulting in a larger sharing cascade.

Finally, we study news sharing and diffusion in populations exhibiting belief polarization. We offer theoretical insights into how the intensity of polarization and the level of perspective diversity within each ideology group (in-group diversity) may facilitate spillovers of less credible news and affect the limit on network connectivity.

We find that increased polarization *decreases* the limit on network connectivity level above which lower credibility news becomes more viral than fully credible news. This is due to the fact that a large fraction of agents who hold extremist beliefs will still share the news after hearing the news with lower credibility. Despite the conventional wisdom that diversifying opinions across the ideological spectrum may mitigate the spread of less credible information in highly-polarized populations, our analysis shows that the impact of increased belief diversity is not necessarily monotone and can even result in wider spread of less credible news: When there is low in-group diversity in perspectives, increased diversity heavily reduces the fraction of the extremists who are in line with the ideology of their own group and impedes sharing of less credible news, increasing the network connectivity limit. However, since increased diversity also increases the number of the extremists whose perspectives have the opposite ideological leaning from their group, the aggregate fraction of extremists is instead rising up when the perspectives are too diverse, propelling the sharing of news with lower credibility and reducing the connectivity limit.

1.1. Related literature

Our work joins the recent line of theoretical research on spread of misinformation. Nguyen et al. (2012); Budak et al. (2011); Törnberg (2018) computationally investigated misinformation diffusion using mechanic models with exogenous rules of relaying information to others. Different from their approaches, we consider Bayesian individuals and formulate news sharing as a strategic and utility-maximizing decision.

Several papers examining phenomena on social media also employed models with rational individuals, grounded in the aforementioned motivations for content sharing or production (Abreu and Jeon, 2019; Acemoglu et al., 2024; Filippas and Horton, 2021; Guriev et al., 2023). Our particular focus on strategically persuading followers to have beliefs similar to one's own through news sharing closely relates to Che and Kartik (2009), in which a sender aims to align the receiver's opinion with her own. This paper then connects to the seminal work in strategic communication: disclosure games (Milgrom, 1981; Grossman, 1981); cheap talk (Crawford and Sobel, 1982); and persuasion games (Milgrom and Roberts, 1986). In our model, agents cannot distort the received news or send additional messages; they only choose whether or not to share news they hear (e.g., by retweeting on X).

A growing body of research investigated the prevalence of misinformation dissemination arising among strategic individuals, often assuming their motivation is to share and act on accurate and truthful information. Kranton and McAdams (2024) explored news markets where news consumers cannot fact-check news and speculate on how likely a news producer may report low-quality news due to costly production of high-quality stories. They demonstrated that low-quality stories prevail when markets are highly connected. This mirrors our finding that in densely-connected social networks, news with lower credibility results in greater dissemination. Papanastasiou (2020) considered perfect inspection on an article and studied decisions of costly inspection and sharing for a stream of strategic agents who have heterogeneous beliefs and cannot observe any inspection actions (nor the results) but only sharing decisions by previous agents. Papanastasiou (2020) posited that more sharing of news by their peers reduced perceived likelihood of fake news, leading to further sharing and news cascade. In contrast, our model incorporates observable news credibility and features strategic substitute in sharing decisions; greater peer sharing only diminishes an agent's incentive to share. Merlino et al. (2023) also

explored a model where individuals could attempt to inspect rumors and the probability of successful inspection increases in effort the individual chooses. They found that truth never circulated alone in social networks; rumors always coexisted. Our work reveals that even fully credible news may fail to spread at all if its content is not extreme.

Acemoglu et al. (2024) presented a closely related study investigating the fact-checking and news-sharing behavior of agents on social networks. Their framework assumed that agents derive utility from future shares of the news they shared while having a reputation concern of being caught sharing misinformation, leading to strategic complementarities in agents' sharing decisions. However, our model differs in its assumptions about the underlying motivations for news sharing, suggesting distinct mechanisms of news sharing. First, our model posits that an informed agent, motivated by persuasion, may share low-credibility news if the agent has a strong incentive and perceives sufficient persuasive value in the news. In contrast, in the work of Acemoglu et al. (2024), agents avoiding spreading misinformation strictly prefer to share the news that they believe is more likely to be truthful. Second, our work explicitly excludes the option of costly fact-check by agents and reputation concern. Consequently, while Acemoglu et al. (2024) showed that increased polarization could lead to fact-checking pressure by the opposite group and thereby curb the spread of misinformation, our model suggests that strong polarization expands the pool of extremists, who are more likely to share news, potentially amplifying spillovers of less credible news.

Moreover, instead of fact-check news to learn its validity, individuals in our model acknowledge that the news is generated among the noisy process of identifying the state and incorporate news value into their beliefs using the known news credibility on a continuous scale.⁵ In this regard, we share the similar approach to investigating the circulating information on platforms with Allon et al. (2021); they studied the evolution of beliefs of consumers who selectively consume only few posts each time when the platforms present a menu of posts that are informative about the true state.

This paper significantly generalizes our preliminary results on news sharing, reported in Hsu et al. (2021). First, unlike the full focus of Hsu et al. (2021) on Gaussian distributions and news that is not fully credible, the present work allows for non-parametric distributions over agents' ex-ante expectations about the unobservable state and characterizes the equilibrium for any news credibility level. This generalized model enables us to quantify the impact of different population characteristics such as polarization and in-group diversity on the size of sharing cascades for different levels of credibility, and in turn on the levels of connectivity required for less credible news to trigger a larger cascade. We deliver new insights into the mechanisms through which news, network connectivity, and beliefs in a population affect individual news-sharing decisions and spread size of news. Specifically, the main contribution of Hsu et al. (2021) was to determine the set of news values that can trigger a cascade, i.e., whether or not the news spreads to a positive fraction of the population (irrespective of the cascade size), and to identify the sufficient and necessary condition under which the spread size is increasing in credibility. By contrast, this paper shows the comparative statics on spread size and characterizes the sufficient and necessary condition under which less credible news can spread to a larger fraction of the population. Moreover, in this paper we uncover the relationship between polarization, in-group diversity, and network connectivity limits, shedding light on spread of less credible news in polarized societies. As a particular case of interest, whereas the specific setting in Hsu et al. (2021) (with no polarization) would suggest a lower connectivity limit if in-group diversity is increased, our new result demonstrates that the effect of in-group diversity can in fact be non-monotone, or even reversed, in a polarized population.

Our work also connects to the theories on information diffusion (Bala and Goyal, 1998; Watts, 2002; Jackson, 2008; Acemoglu et al., 2010; Sadler, 2020), as well as the work on information spillover via word-of-mouth (Campbell, 2013; Campbell et al., 2017; Ajorlou et al., 2018). Our focus on individuals' strategic news-sharing decisions on social networks is also related to the recent literature on strategic interactions in large random graphs (Parise and Ozdaglar, 2021; Board and Meyer-ter Vehn, 2021).

Our study on news spread within polarized populations contributes to the broader literature on polarization in societies (DiMaggio et al., 1996; Fiorina et al., 2005; Baldassarri and Gelman, 2008; Abramowitz, 2010). Relevant to this is the study of Guess et al. (2018), who found partisan differences in misinformation consumption on social media during the 2016 U.S. presidential campaign, and suggested that social media platforms might exacerbate selective exposure to fake news. Complementing this, our work examines news-sharing behavior when a population is polarized into two belief camps, offering a theoretical perspective on how polarization influences information spread.

The rest of the paper proceeds as follows. We describe our model and discuss the assumptions in Section 2. We analyze the equilibrium of our model in Section 3, elucidating the interaction between sharing decisions and scale of news spread. In Section 4 we identify conditions for lower credibility news to trigger a larger cascade than fully credible news, using which we characterize the limit on network connectivity. We then provide the implications of our model for news spread in populations with polarized perspectives. We conclude in Section 6. The technical proofs of all lemmas and theorems are all relegated to the Appendix.

2. Model

We consider a unit-mass continuum of agents, each indexed by a real number $i \in I \triangleq [0, 1]$. Social interactions are characterized by a directed network $\mathcal{G} = (I, \mathcal{E})$. The population is concerned with an unobservable state $\theta \in \mathbb{R}$, against which each agent will take a binary action in a later stage. Ex ante the agents have heterogeneous subjective prior beliefs about the state. Each agent, before

⁵ In our previous work (Hsu et al., 2020a), each informed agent does not inspect news and updates her belief by placing real-valued weights on the news and her own prior mean according to the news generation process. In contrast to the present paper, Hsu et al. (2020a) assumed that an agent desires to make her followers beliefs close to hers according to some distance measure on the space of beliefs, instead of aligned binary actions. Also, Hsu et al. (2020a) assumed that informed agents make myopic sharing decisions, speculating the probability that their followers may have received the news as of the time they are making a sharing decision, without projecting on the scale of spread at its steady state.

taking a binary action, interacts with her social contacts and, if she observes a piece of news, may share the news with her followers, influencing her followers' beliefs about the state.

The game proceeds in three stages 0, 1, 2. In stage 0, Nature makes three moves: (i) Each agent is independently assigned a prior belief about the state, drawn from a common distribution; (ii) the social network $\mathcal{G}$ is generated based on a simple random network model; (iii) news is generated and given to some random agents with infinitesimal size of the population. All agents are initially unaware of the existence of the news except for the agents who received the news from Nature. In stage 1 agents interact: Upon seeing some news, an agent has to decide immediately whether to broadcast the news or not. We also call stage 1 as interaction stage. Stage 2 is the action stage where each agent simultaneously takes a binary action. The utilities resulting from sharing decisions and actions are realized after the action stage. We provide the specifics as below.

Network of social interactions. The directed network $\mathcal{G} = (\mathcal{I}, \mathcal{E})$ is exogenously constructed, representing follower–followee relationships among the agents. We say that agent j follows agent i if and only if $(j, i) \in \mathcal{E}$: In this relationship, agent j is agent i 's follower and agent i is agent j 's followee. For each agent i , we denote the set of her followers as $N_i^{\text{in}} \triangleq \{j \in \mathcal{I} | (j, i) \in \mathcal{E}\}$ and let $\ell_i \triangleq |N_i^{\text{in}}|$ be the cardinality of this set, to which we refer as agent i 's in-degree. Similarly, we write $N_i^{\text{out}} \triangleq \{j \in \mathcal{I} | (i, j) \in \mathcal{E}\}$ for the set of agent i 's followees and let $d_i \triangleq |N_i^{\text{out}}|$ denote agent i 's out-degree.

Network $\mathcal{G}$ is constructed as follows. Each agent i is randomly assigned with two integers as her in-degree ℓ_i and out-degree d_i respectively, where her in-degree ℓ_i is drawn from an in-degree distribution $P^{\text{in}}(\cdot)$ and out-degree d_i is drawn from an out-degree distribution $P^{\text{out}}(\cdot)$. We assume that the in-degrees and out-degrees of agents are independent of each other and both in-degree and out-degree distributions are identical to a common discrete degree measure $P^{\text{deg}}(\cdot)$.⁶ With this assumption, we say that the agents with in-degree ℓ (out-degree d) account for the fraction $P^{\text{deg}}(\ell)$ ($P^{\text{deg}}(d)$) of the population by informal use of the law of large numbers. Agent i with out-degree d_i is then randomly assigned with d_i others according to an atomless weighted-uniform distribution on the population $[0, 1]$, where the weights are proportional to the in-degrees of the agents.⁷

Throughout this paper we make a particular choice for the degree distribution:

Assumption 1. We consider Poisson degree distribution with parameter λ , i.e., $P^{\text{deg}}(d) = \frac{\lambda^d}{d!} e^{-\lambda}$, $d = 0, 1, 2, \dots$. The parameter λ is the expected degree of Poisson distribution to which we refer as the network connectivity. We focus on the range $\lambda > 1$ for which there exists at least one weakly-connected component in the network comprising a positive fraction of the population.⁸

The abstract network model is common knowledge to all the agents while the realization $\mathcal{G}$ is not: Agents have no knowledge about the interactions among others and only know their local positions in the network $\mathcal{G}$. Specifically, each agent only knows the sets of her followers N_i^{in} and followees N_i^{out} , which are her private information.

Prior beliefs and types of agents. The agents have heterogeneous subjective prior beliefs about the state θ . We assume that their prior beliefs only differ with their real-valued prior expectations (means) about the state. Individual prior means are their private information; we denote each agent i 's prior mean about the state as $\mu_i \in \mathbb{R}$, and call it agent i 's *perspective* (or type). However, it is commonly known that each agent's perspective is independent and identically distributed according to a common CDF F , i.e., for any agent i

$$\mu_i \sim F.$$

Assumption 2. We assume that CDF F is continuously differentiable with corresponding density function f and full support on the real line. The mean and variance of function f are finite. For simplicity of exposition, we assume that $F(0) = \frac{1}{2}$. We write $\mathcal{F}$ for the space of the CDFs.

News, credibility, and belief updating. Nature generates a single piece of news $x \in \mathbb{R}$, informative about the state and includes the news source in the news so that agents can distinguish copies of the same news. We assume that all agents agree on the precision of the news source; the objective precision is measured as the *credibility* $\beta \in [0, 1]$, which quantifies the sensitivity of agents' beliefs in reaction to the news.

When an agent i receives the news (x, β) , she then knows that the news is circulating in the population and updates her belief about the state. We denote as $z_i \in \{\emptyset, (x, \beta)\}$ her private history of being informed or not: We set $z_i = \emptyset$ when agent i has not received any news while $z_i = (x, \beta)$ if she has received the news. We abstract the belief updating process by focusing only on each agent i 's belief mean, which we denote as $\mathbb{E}_i[\theta | \mu_i, z_i]$ given her type (μ_i) and whether she is informed or not (z_i). When agent i has not received

⁶ Hsu et al. (2020a) analyzed the news spread using the configuration model for the construction of random networks, considering a general joint degree distribution in the population.

⁷ Specifically, an agent with in-degree ℓ is ℓ times likely to be drawn as a followee than an agent with in-degree 1, i.e., the probability density that agent i is sampled is $\frac{\ell}{\mathbb{E}[\ell]}$ where $\mathbb{E}[\ell]$ is the average degree. See Galeotti and Goyal (2009); Fainmesser and Galeotti (2016, 2020) for this modeling approach.

⁸ Many empirical studies indeed support large average degrees of social networks Java et al. (2007); Mislove et al. (2007); Kwak et al. (2010); Zhou et al. (2010) and hence $\lambda \leq 1$ is not of practical interest. For example, the dataset which was collected and kindly shared by Kwak et al. (2010) for the follower network on Twitter (as the motivating application of the present work) comprises over 41.7 million nodes and 1.47 billion links, with a value of 35.2 for the average degree.

the news ($z_i = \emptyset$) and hence stays unaware of the news, her belief mean stays unchanged, i.e., $\mathbb{E}_i[\theta|\mu_i, \emptyset] = \mu_i$. For an agent who has observed the news, we particularly consider the following linear updating rule for the mean of her belief⁹:

Assumption 3. For each agent i , if she observes the news (x, β) , her belief mean is updated as

$$\mathbb{E}_i[\theta|\mu_i, (x, \beta)] = \beta x + (1 - \beta)\mu_i, \quad (1)$$

where the credibility β is the sensitivity of the updated mean to the observed news.¹⁰

Sharing decision, action, and utility function. The interaction stage takes place in discrete time $t = \{0\} \cup \mathbb{N}$. At $t = 0$, news (x, β) is released to a subset of the population with an infinitesimal mass $\delta > 0$. This subset's agents are drawn randomly and uniformly from the population, independent from their network positions and perspectives. Upon receiving news at time t , agent i must make a binary sharing decision $s_i \in \{0, 1\}$ at t as whether to broadcast the news to their followers or not. If agent i shares the news at t (i.e., $s_i = 1$), all of agent i 's followers receive and observe the news at $t + 1$. Sharing news, however, incurs a cost, which will be specified in the utility functions.

We, following the approach of Sadler (2020), assume that upon observing the news, agents are unaware of when the news spread started: They know the stages of the game, but do not know the time corresponding to $t = 0$ and the time of receiving the news. Therefore, agent i 's sharing strategy is independent of time t , denoted by $s_i(\mu_i, z_i, F) : \mathbb{R} \times \{\emptyset, (x, \beta)\} \times F \rightarrow \{0, 1\}$, where $s_i(\mu_i, \emptyset, F) = 0$ because she can share news only if she receives one. The inclusion of F as an argument captures agent i 's consideration of her followers' perspectives on the state in her decision making. By convention, we write s as the profile of sharing strategies; we denote by $s((x, \beta), F)$ the sharing strategy profile for news (x, β) and F .

We assume that the interaction stage is long enough that the news spread reaches its steady state.¹¹ We then define the size of news spread as follows:¹²

Definition 1. Given λ , F , sharing strategy profile s , and the initial seeding size δ , the δ -spread size for news x with credibility β is the fraction of the population who have received the news at the steady state and we denote it by $q(\lambda, F, s((x, \beta), F), \delta)$.

In the action stage, each agent i takes a binary action $a_i \in \{-1, +1\}$ that matches the sign of her belief mean. Specifically, based on the belief updating (1), agent i takes an action using the following rule¹³:

$$a_i(\mu_i, z_i) = \text{sign}(\mathbb{E}_i[\theta|\mu_i, z_i]). \quad (2)$$

Let a denote the profile of the agents' actions.

After the action stage ends, agent i 's utility resulting from her sharing decision s_i and the agents' actions a is realized, and is given by

$$u_i(\theta, s_i, a) = \theta(w^I a_i + w^L \sum_{k \in N_i^{\text{in}}} a_k) - \left| N_i^{\text{in}} \right| C \times \mathbb{1}_{\{s_i=1\}}, \quad (3)$$

the weights $w^I, w^L \in (0, 1)$ are common for all the agents, satisfying $w^I + w^L = 1$, and $C > 0$ represents a universal average cost incurred per follower when broadcasting news. The utility function (3) captures an agent's preference for alignment of collective actions with the state at individual and local scales, comprising a convex combination of two components: (1) agent i 's individual action a_i ; (2) the aggregate actions of agent i 's followers. Agent i derives positive utility when the weighted average of collective

⁹ This particular updating rule for the mean of one's belief can be rationalized by considering a distribution in the exponential family for news generation and the associated conjugate prior as an agent's prior on the state (Diaconis and Ylvisaker, 1979). For example, consider a Gaussian belief formation process: An agent i has a Gaussian prior belief on the state centered at her perspective, i.e., $\theta \sim_i \mathcal{N}(\mu_i, \sigma_\theta^2)$ for some variance $\sigma_\theta^2 > 0$. News item is an observation about the state with an additive independent Gaussian noise, i.e., $x = \theta + \epsilon$ where $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ for some $\sigma_\epsilon^2 > 0$. When agent i observes news x and updates her belief on the state using Bayes' rule, one can show that her posterior belief on the state is still a Gaussian variable, characterized by $\theta|x \sim_i \mathcal{N}(\frac{\sigma_\epsilon^2}{\sigma_\theta^2 + \sigma_\epsilon^2} \mu_i + \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\epsilon^2} x, \frac{\sigma_\theta^2 \sigma_\epsilon^2}{\sigma_\theta^2 + \sigma_\epsilon^2})$ where the posterior mean is a convex combination of her perspective μ_i , the news x and the weights are controlled by the belief uncertainty and noise variance.

¹⁰ In a more general setting, credibility of news can be subjective, varying with agents' own perspective and news value. We will discuss this setting in Section 3 where we characterize the structure of equilibria.

¹¹ Empirical studies tracking online trends and social media activity (Leskovec et al., 2009; Schema et al., 2019) have indicated that the lifespan of online news is typically less than two weeks. This is a relatively short duration compared to sustained behavioral actions, such as mask-wearing.

¹² The present work concerns the size of news-sharing cascades arising from individuals' sharing decisions, with no reference to structural properties of news-sharing dynamics on online social networks (see Vosoughi et al. (2018); Zhao et al. (2020)). This choice enables tractable analysis with focus on the individuals' motivation for sharing news, and is also consistent with the findings by Juul and Ugander (2021) that the structural difference in news spread can be largely explained by difference in spread size, and moreover, by the transmission rate of news between peers.

¹³ For simplicity we assume that agents take the action $+1$ when their belief mean is zero. This simplification does not affect agents' strategies of sharing in the following analysis as we consider a continuum of agents and the measure of such agents is zero.

actions matches the sign of the state and the gain/loss is increasing in the magnitude of state. Since the agents have different subjective beliefs on the state, agent i has an incentive to influence, via sharing the news, her followers' beliefs and in turn their actions.¹⁴

We solve for Bayesian Nash equilibria (BNE) of the game, characterizing the news-sharing strategies of the agents. Note that we exclude from each agent i 's strategy formulation the private information about her followees and followers $N_i^{\text{in}}, N_i^{\text{out}}$. Indeed, in our model the strategies are independent of the identities of one's followees and followers as well as her in-degree and out-degree, as we will demonstrate in Section 3.

Our particular focus is on whether news can spread to a non-zero fraction of the population at steady state even when the size of the initial seeding δ is infinitesimal.

Definition 2. Given λ, F , and strategy profile s , we say that a cascade of news (x, β) emerges almost surely if and only if $\lim_{\delta \rightarrow 0} q(\lambda, F, s((x, \beta), F), \delta)$ exists and is greater than 0. We abusively denote the limit as $q(\lambda, F, s((x, \beta), F))$ to which we refer as the corresponding spread size (or cascade size).

Discussion of the model and assumptions

Being unaware of news. Many models concern the players who are unaware of products, opportunities, or rumors until they observe their neighbors adopted or communicated.¹⁵ Sadler (2020) studied a diffusion game in which strategic players who are informed of the opportunity via their neighbors' adoption decide whether to adopt it as well considering network externality. Merlino et al. (2023) investigated the diffusion of rumors among strategic agents who can fact-check the rumor once they receive the rumor and know its existence.

However, if an agent speculates some news may exist and has not yet received it, she strategically interprets not hearing news as possibly a result of her followees deciding not to share it. This level of strategic speculation primarily concerns niche applications in which the access to information may be costly or restricted to some players, such as strategic communication at private meetings or organizations. Hsu et al. (2020b) investigated news subscription choices of fully rational subscribers who actively seek information, accounting for media bias in news disclosure. In the context of news sharing on Twitter-like platforms where users are passively exposed to many tweets on a wide range of topics, such level of strategic speculation for the existence of specific news is not realistic. Moreover, if agents speculate that relevant news is circulating in the society, they can readily look for it via digital search engines.

Local persuasion motives. Our model focuses on local persuasion motives, where individuals are primarily motivated to influence their immediate social circles rather than the entire population. This assumption is well-grounded in behavioral and empirical studies on information diffusion, which show that people prioritize the opinions and actions of their close connections—such as friends, family, or direct followers—over broader societal trends (Campbell et al., 2017; Sadler, 2020; Acemoglu et al., 2024). In contrast, entities such as digital advertisers, media outlets, and political actors/campaigns often adopt a macroscopic perspective, strategizing for population-wide outcomes.¹⁶

Assured reception and observation of shared news by followers. Several factors may prevent followers from observing shared news. The ranking algorithm deployed by social media platforms may not prioritize the content for followers' feeds. Furthermore, limited human attention, as described by Kahneman (1973), poses a challenge. Especially in the contemporary era that is inundated with readily accessible information, attention has become a scarce resource (Simon, 1996; Davenport and Beck, 2001). Abreu and Jeon (2019) incorporated this attention factor into individuals' news-sharing decisions, considering whether the potential benefit their followers derive from reading the news can exceed the cost of their attention. To focus on the persuasive motive, our model simplifies by not accounting for this possibility that a follower may not observe shared news.

News shared with perfect fidelity. This model assumes perfect fidelity in news sharing: agents cannot alter news, and all recipients observe the same content and credibility. This allows agents to predict the persuasiveness and spread of news (see Section 3). However, on social media platforms, digital information can mutate during transmission (Adamic et al., 2016), either intentionally (Fallis, 2009) or unintentionally due to misinterpretation (Simmons et al., 2011). Our model simplifies this factor, capturing scenarios like retweeting an article where the original content remains intact.¹⁷

¹⁴ An alternative utility function, used in an earlier version of our work, is

$$u_i(\theta, s_i, a) = \theta(w^I a_i + w^L \frac{1}{|N_i^{\text{in}}|} \sum_{k \in N_i^{\text{in}}} a_k) - C \times \mathbb{1}_{\{s_i=1\}}, \quad (4)$$

where neither the utility from followers' actions nor the cost of sharing scales with the number of followers. As we elaborate later (in footnote or remark), both formulations lead to identical equilibrium sharing decisions and actions.

¹⁵ One can see Bass (1969); Campbell (2013) for diffusion based on non-strategic information transmission.

¹⁶ Abreu and Jeon (2019) and Kranton and McAdams (2024), for instance, analyzed news suppliers seeking to maximize advertising revenue derived from overall user engagement. Similarly, Hsu et al. (2022) explored the strategic disclosure and verification practices of news media that focus on the welfare of the population and influence readers to take their preferred action.

¹⁷ Jackson et al. (2022) explored structural interventions on social media networks that can mitigate false communication, when information can be distorted during transmission.

News sharing without temporal considerations. Time can be a critical factor in information-sharing decisions. Individuals may have stronger incentives to share information early in its dissemination if they specifically seek to be the ones who influence their followers' opinions; such motivations may include the desire to appear knowledgeable Leary and Kowalski (1990); Baumeister (2010), to establish themselves as opinion leaders Rogers (2003), or to gain recognition from peers Wang and Fussell (2020). Similarly, the cost of sharing associated with risk of toxic discussion or harassment can also vary with time, as early sharing may attract greater user attention and engagement. The assumption of agents' unawareness of time in the dissemination process (same approach as in Sadler (2020)) abstracts away from these time-dependent incentives, enabling a tractable analysis of time-invariant sharing strategies.

Cost of sharing. Our model posits that the cost associated with news sharing is proportional to the number of followers, reflecting the increased likelihood of online harassment or harmful commentary with larger audiences. The cost of sharing may otherwise vary with factors such as news value, credibility, and agents' perspectives—potentially decreases for news aligned with an agent's perspective or highly credible news. To avoid introducing bias, we assume a fixed sharing cost per follower in this model. Exploring these cost variations is a direction for future research.

We use the following terminology when discussing our results. The binary actions to match the sign of an unobservable state naturally splits the information space into three ideological leanings. The agents with positive perspective ($\mu_i > 0$) are *right-leaning*, the agents with negative perspective ($\mu_i < 0$) are *left-leaning*, and the agents with zero perspective ($\mu_i = 0$) are neutral. News items are similarly categorized according to their value: *right-leaning* ($x > 0$), *left-leaning* ($x < 0$), and *neutral* ($x = 0$). The absolute value of news x , i.e., $|x|$, is the *news magnitude*, which measures how extreme the news is away from the neutral point. Finally, we say that news is fully credible when it has the full level of credibility, i.e., $\beta = 1$.

The action space $\{-1, +1\}$ naturally splits the space of real numbers into three ideological leanings: *right-leaning* individuals with positive prior mean $\mu_i > 0$, who prefer action $+1$ ex-ante, *left-leaning* individuals with negative prior mean $\mu_i < 0$, who prefer action -1 ex-ante, and *neutral* individuals with $\mu_i = 0$, who are ex-ante indifferent; note that the measure of neutral subscribers is zero. We say that a subscriber j exhibits *homophily* if she subscribes to the intermediary who shares his ideological leaning ex-ante, i.e., $\text{sign}(\mu_j) = \text{sign}(\mu_s)$; we say that she exhibits *anti-homophily* if he subscribes to the one with the opposite ideological leaning. Moreover, we say that j exhibits weak (resp. strict) homophily if she weakly (resp. strictly) prefers the intermediary who shares his own ideological leaning ex-ante. Weak and strict anti-homophily are defined similarly.

3. Equilibrium analysis

We first derive the mechanics of how news spread emerges from the agents' news-sharing decisions. We then elucidate on agents' decision making given their knowledge about the network structure and speculation on news spread, deriving their equilibrium news-sharing strategies.

3.1. News spread emerging from news-sharing decisions

Based on the random network model and the fact that individual perspectives are private information, we compute the δ -spread size for news x at the steady state using susceptible-infection model (SI model, see Jackson (2008)), the network structure (Poisson network with connectivity λ), and the fraction of informed agents who share the news.

Let $P_F(s((x, \beta), F))$ denote the fraction of informed agents who share the news (x, β) according to sharing strategy profile s , measured with respect to perspective CDF F ¹⁸:

$$P_F(s((x, \beta), F)) \triangleq \mathbb{P}(s_i(\mu_i, (x, \beta), F) = 1 | i \in \mathcal{I}, \mu_i \sim F).$$

Since each following link is drawn uniformly at random from the population, the probability that at the steady state a random following link points to an agent who has shared the news equals the probability that it follows an agent who has received the news (q) and would like to forward the news (P_F). At the steady state, the fraction of the agents who has not received the news at the whole interaction stage is $1 - q$, which are the agents who did not receive it at $t = 0$ ($1 - \delta$) and has not received it from any of her followees. We can hence write the equation for the probability that a random agent does not hear the news at the steady state:

$$1 - q = (1 - \delta) \sum_{d=0}^{\infty} P^{\text{out}}(d) (1 - q P_F(s((x, \beta), F)))^d.$$

Using $P^{\text{out}}(\cdot) \sim \text{Poisson}(\lambda)$ and taking $\delta \rightarrow 0$, we obtain the following result:

Lemma 1. Given λ, F , and sharing strategy profile s , the spread size $q(\lambda, F, s)$ for news x with credibility β satisfies

$$q = 1 - e^{-q\lambda P_F(s((x, \beta), F))}. \quad (5)$$

¹⁸ Since we assume independence of individual perspectives and agents broadcast the news instead of forwarding the news to a subset of her followers they select, receiving the news is independent from the agent's perspective. Therefore, the chance of sharing news for an agent who receives the news equals the fraction of the population who will share the news upon seeing it.

Moreover, $q = 0$ if $\lambda P_F(s((x, \beta), F)) \leq 1$.¹⁹

Different from the conventional version of SI model in which the probability of information transmission is an exogenous constant, in our model the probability of news sharing is a function of sharing strategy profile s , which endogenously hinges on the size of news spread, as will be shown later.

3.2. News-sharing decisions in light of news spread

We adopt the conventional notation s_{-i} for the profile of sharing strategies other than i . Upon receiving the news and observing its credibility (x, β) , agent i 's sharing decision $s_i \in \{0, 1\}$ is her best response to agents' action rule a and others' sharing strategies s_{-i} , if and only if there is no deviation strictly improving her expected utility:

$$\mathbb{E}_i \left[u_i(\theta, s_i, a) | s_{-i}, \mu_i, z_i = (x, \beta) \right] \geq \mathbb{E}_i \left[u_i(\theta, 1 - s_i, a) | s_{-i}, \mu_i, z_i = (x, \beta) \right], \quad (6)$$

where the expectation is taken over agent i 's belief about all the other agents' perspectives on the state and their private histories at the action stage. To evaluate (6), agent i needs to speculate on the influence of her sharing decision s_i , as well as other agents' sharing decisions, on any agent's private history. However, it suffices to consider only whether her followers are informed or not. In a population with a continuum of agents, the sharing decision of a single agent with a finite number of followers has a negligible effect on the spread size of news. Therefore, given any s_{-i} , agent i 's sharing decision only influences the beliefs and actions of her followers, without affecting the spread size; equivalently, agent i evaluates whether persuading her followers is worthy of the cost of sharing.

Specifically, choosing to broadcast the news ($s_i = 1$) ensures that all of her followers will hear the news. On the other hand, if she decides not to broadcast ($s_i = 0$), she has to weigh in the chance that some of her followers may end up hearing the news from other agents at the steady state of the news spread. For Poisson networks, one can show that this chance equals the δ -spread size of the news:

Lemma 2. *On a Poisson network with connectivity λ , given δ, F , and profile of sharing strategies s , for any follower of agent i , say $k \in N_i^{\text{in}}$, the probability that follower k receives the news (x, β) from agents other than agent i at the steady state of the news spread is equal to the δ -spread size of the news.²⁰*

Therefore, when δ is infinitesimal, for any agent i 's follower $k \in N_i^{\text{in}}$, we can derive the probability that follower k is informed at the steady state of the news spread conditional on agent i 's sharing decision s_i and other agents' sharing strategies s_{-i} as²¹:

$$\mathbb{P}(z_k = (x, \beta) | s_i, s_{-i}) = \begin{cases} 1, & \text{if } s_i = 1; \\ q(\lambda, F, s((x, \beta), F)), & \text{if } s_i = 0. \end{cases} \quad (7)$$

The aggregate actions of agent i 's followers can then be computed as

$$\begin{aligned} & \mathbb{E}_i \left[\sum_{k \in N_i^{\text{in}}} a_k(\mu_k, z_k) | s_i, s_{-i}, z_i = (x, \beta) \right] \\ &= \begin{cases} |N_i^{\text{in}}| \mathbb{E}_i [a_k | k \in N_i^{\text{in}}, z_k = (x, \beta)], & \text{when } s_i = 1; \\ q |N_i^{\text{in}}| \mathbb{E}_i [a_k | k \in N_i^{\text{in}}, z_k = (x, \beta)] + (1 - q) |N_i^{\text{in}}| \mathbb{E}_i [a_k | k \in N_i^{\text{in}}, z_k = \emptyset], & \text{when } s_i = 0. \end{cases} \end{aligned} \quad (8)$$

By substituting (6) and removing the utility derived from one's own action, we deduce that an informed agent strictly prefers to share the news if and only if:

$$\mathbb{E}_i [\theta | \mu_i, (x, \beta)] (1 - q) |N_i^{\text{in}}| \left(\mathbb{E}_i [a_k | k \in N_i^{\text{in}}, z_k = (x, \beta)] - \mathbb{E}_i [a_k | k \in N_i^{\text{in}}, z_k = \emptyset] \right) > |N_i^{\text{in}}| C. \quad (9)$$

As individual perspectives are private information and each follower k uses the same action rule (cf. (2)), we observe from (9) that all informed agents are faced with the same evaluation process, independent of the identity and number of their own followers (which

¹⁹ It can be shown that $q = 0$ is the only solution to (5) if and only if $\lambda P_F(s((x, \beta))) \leq 1$. When $\lambda P_F(s((x, \beta))) > 1$, the spread size is the non-zero solution to (5) since $q = 0$ is not a stable solution due to the infinitesimal fraction of the population who initially received the news. See the proof of Theorem 1 in the Appendix and Jackson (2008) for more details.

²⁰ For general out-degree distributions, the identity does not necessarily hold due to two factors: (i) the out-degree distribution of a follower should be adjusted by the well-known friendship paradox; (ii) the chance is evaluated conditioned on the event that a follower with out-degree d can receive the news only from any of her other $d - 1$ followees.

²¹ Given any s_{-i} the value $q(\lambda, F, s_i = 0, s_{-i})$ in (7) is independent of s_i because the sharing decision of a single agent does not affect the spread size of news in a continuum of agents.

merely scales the benefit/cost of sharing), and consequently deploy the same sharing strategy in the game.²² We henceforth refer to s as the common sharing strategy.

The following lemma characterizes an agent's decision rule in news sharing:

Lemma 3. *Given x, β, λ, F , and the endogenous spread size $q(\lambda, F, s((x, \beta), F))$ arising from the strategy s , agent i strictly prefers to broadcast the news if the following inequality holds:*

$$\underbrace{\left[\beta x + (1 - \beta) \mu_i \right]}_{\text{agent } i \text{'s belief mean}} \underbrace{\left[2(1 - q) F^{\text{flip}}(x, \beta) \right]}_{\text{effective persuasive power of the news}} > \frac{C}{w^L}, \quad (10)$$

where

$$F^{\text{flip}}(x, \beta) \triangleq F(0) - F\left(\frac{-\beta x}{1 - \beta}\right),$$

measures the fraction of agents who will change their favored actions upon receiving the news (x, β) . Agent i does not broadcast the news if the reversed inequality of (10) holds.²³

Three elements are identified as indicated in (10): belief mean, effective persuasive power of the news, and the adjusted cost of sharing.

The effective persuasive power of the news measures the probability that a random follower will flip her favored action *directly due to* the agent's sharing. Specifically, the sign of $F^{\text{flip}}(x, \beta)$ indicates the direction of action flipping, which is determined by the sign of news: Right-leaning (resp. left-leaning) news can only change the favored action from -1 to $+1$ (resp. $+1$ to -1). The magnitude, i.e., $\left| F(0) - F\left(\frac{-\beta x}{1 - \beta}\right) \right|$ is the fraction of agents who flip their favored actions when seeing news x . However, considering the probability that a follower may also receive the news from other agents, this difference has to be discounted by $1 - q$ to reflect the effective fraction that can be persuaded *only* by the decision maker, showing the strategic substitutability of news-sharing decisions. Finally, the multiplicity 2 accounts for the change in aggregating actions when one agent flips her favored action.

We observe that the magnitude of persuasive power is increasing in both news magnitude and credibility: Farther the news is away from being neutral or higher the news credibility is, then a larger fraction of the population will switch their favored actions upon seeing the news. In particular, when $\beta = 0$, no agents share the news since the news has zero persuasive power, generating no benefits of sharing. It is a trivial scenario where news does not spread; for simplicity we hereafter exclude $\beta = 0$ from our discussion.

The adjusted cost reflects an agent's concern about the aggregate actions of her followers in her sharing decision: A larger weight w^L means a greater level of concern, increasing her willingness to persuade her followers, or equivalently, diminishing her fear of online harassment or negative discourse. For simplicity of exposition, we henceforth use notation C to represent the adjusted cost, normalized with the weight w^L .

In (10) the product of belief mean and the effective persuasive power is the gain from sharing the news, exhibiting monotonicity in perspective given all the other parameters. Agents with more extreme perspectives place higher value on their actions (absolute value of their belief means), and hence have stronger incentives to share or not to share the news, depending on whether the sign of news is in line with their beliefs. This observation suggests a structural form for sharing strategy that we characterize in the following definition:

Definition 3. We say that agent i deploys a threshold sharing strategy if given F and β , there exists a set of thresholds $\mu^{\text{th}}((x, \beta), F)$ for each x such that

$$s(\mu_i, (x, \beta), F) = \begin{cases} \mathbb{1}\{\mu_i > \mu^{\text{th}}((x, \beta), F)\}, & \text{when } x \geq 0; \\ \mathbb{1}\{\mu_i < \mu^{\text{th}}((x, \beta), F)\}, & \text{when } x < 0. \end{cases} \quad (11)$$

where $\mathbb{1}\{\cdot\}$ denotes the indicator function.

It is noteworthy that the presence of sharing cost introduces strategic interaction into this model. If the cost vanishes to zero, then from (10), the effect of spread size on an informed agent's sharing decision disappears, as $1 - q$ is positive. This implies that without the sharing cost, sharing decisions reduce to a straightforward assessment of belief alignment with follower action flipping, eliminating strategic substitutes.

²² Agents with no followers (i.e., $|N_i^{\text{in}}| = 0$) are indifferent between sharing or not sharing and their sharing decisions do not affect the news spread; we assume they follow the same sharing strategy as other agents. Furthermore, as the set of followees provide no additional information and are not involved in the utility function, each agent's sharing decision is independent of N_i^{out} .

²³ Sharing decisions remain identical if agents alternatively make their sharing decisions based on the average utility/cost of sharing over their followers as stated in (4) (vis-à-vis the total utility/cost formulated in (3)). This follows from the observation that the utility of one's own action cancels out when comparing expected utilities between sharing or not sharing, resulting in a normalized version of the inequality presented in (9).

Remark 1. A more general setting is to make the credibility of news subjective, particularly depending on agents' perspectives and news value. An agent may deem news that aligns with her prior belief as more credible, while often disregarding the news that is against her perspective. In such a setting, one can find that, under some mild regularity assumptions on agents' prior beliefs and distribution of news, an agent's belief mean about the state given news is increasing in her perspective.²⁴ As a result of such monotonicity, each agent still deploys a threshold sharing strategy, noting that the effective persuasive power of news in (10) does not vary across the agents. The equilibria for both cases share a qualitatively similar structure. For expositional purposes, however, we study the setting in which agents base their decisions on a common observable news credibility.

3.3. Equilibrium structure

Lemma 1, 2 and 3 together identify the interplay between the scale of news spread at macro-level and individual threshold strategies via the fraction of informed agents who share the news. Noting that the sharing decision rule in Lemma 3 becomes the same for the informed agents when $\beta = 1$, we proceed with characterizing the equilibrium in the two cases of $\beta < 1$ and $\beta = 1$.

Not fully credible news ($\beta < 1$)

Lemma 3 establishes a strategy following the form (11) whereby the threshold μ^{th} must satisfy the indifference equation²⁵

$$\left[\beta x + (1 - \beta) \mu^{\text{th}} \right] \left[2 \left(1 - q(\lambda, F, s((x, \beta), F)) \right) F^{\text{flip}}(x, \beta) \right] = C. \quad (12)$$

The following theorem identifies the equilibrium for $\beta < 1$.

Theorem 1. *Given any $\beta < 1$, for any x, λ, F , an equilibrium exists and it is unique, with the following properties:*

- (i) *The agents deploy a threshold sharing strategy.*
- (ii) *The common threshold $\mu^{\text{th}*}$, the steady-state spread size q^* and the sharing fraction P_F^* are the unique solution of the following system of equations:*

$$\begin{aligned} \mu^{\text{th}*} &= \frac{C}{2(1 - q^*)F^{\text{flip}}(x, \beta)} - \beta x && \text{(indifference condition)} \\ P_F(s^*) &= \begin{cases} 1 - F(\mu^{\text{th}*}) & \text{when } x \geq 0; \\ F(\mu^{\text{th}*}), & \text{when } x < 0, \end{cases} && \text{(sharing fraction)} \\ q^* &= 1 - e^{-q^* \lambda P_F(s^*)}. && \text{(spread dynamics)} \end{aligned}$$

For news that is not fully credible, the beliefs of informed agents about the state are still heterogeneous: The informed agents with extreme beliefs on one ideological wing strictly prefer to share the news whereas those holding extreme beliefs with the opposite ideology strictly prefer not to share. The equilibrium sharing fraction is therefore non-zero.

Fully credible news ($\beta = 1$)

When $\beta = 1$, for the informed agents, the mean of their beliefs are identical and equal to the news value. Lemma 3 hence yields the identical criterion for all the informed agents for news sharing:

$$|x| \left(1 - q(\lambda, F, s((x, 1), F)) \right) \gtrless C. \quad (13)$$

As a result, the informed agents may (i) all strictly prefer sharing (when “ $>$ ” holds in (13)); or (ii) all strictly prefer not sharing (“ $<$ ” holds); or (iii) all feel indifferent (“ $=$ ” holds). Case (i), in which each agent shares the news (i.e., $s \equiv 1$), is an equilibrium if

$$|x| > \frac{C}{1 - q(\lambda, F, s \equiv 1)},$$

where $q(\lambda, F, s \equiv 1)$ is in fact the size of the giant component of network $\mathcal{G}$ when every agent shares the news. Similar for case (ii) where no agent shares the news (i.e., $s \equiv 0$), the spread size is 0, and not sharing is an equilibrium if

$$|x| < C.$$

For the last case (iii), any strategy that assigns sharing to an arbitrary subset of perspectives sustains an equilibrium if the induced fraction of agents who share the news given by $P_F \in [0, 1]$, and the consequent spread size q satisfy

²⁴ See the extensive study of Hsu et al. (2020b), where authors account for subjective credibility and rationalize the belief updating process via an information structure which allows for both informative and uninformative news about the state.

²⁵ The agents whose perspectives equal the threshold feel indifferent between sharing or not; for simplicity we assume that they choose not to share the news. Again this simplification does not affect agents' strategies of sharing as the measure of critical agents is zero. When $x = 0$ the decision rule (10) suggests no sharing by any informed agents, which is reflected by the corresponding threshold of positive infinity in (12).

$$|x| = \frac{C}{1-q}$$

$$q = 1 - e^{-q\lambda P_F}.$$

The equilibria in all three cases are in fact in the class of threshold strategies: The cases (i) and (ii) correspond to a threshold strategy (11) with thresholds of infinite magnitude, while for case (iii) the threshold is finite. The next theorem summarizes the above discussion deriving the closed-form solutions for spread size, the fraction of informed agents sharing news, and the sharing threshold at the equilibrium for $\beta = 1$.

Theorem 2. Suppose $\beta = 1$. Given x, λ, F there exists an equilibrium possessing the following properties:

- (i) The agents deploy a threshold sharing strategy.
- (ii) The common threshold $\mu^{\text{th}*}$, the steady-state spread size q^* and the sharing fraction P_F^* are given by:

$$(q^*, P_F^*)(x) = \begin{cases} (0, 0), & \text{if } |x| \in [0, C]; \\ (1 - \frac{C}{|x|}, \frac{-\ln(\frac{C}{|x|})}{\lambda(1 - \frac{C}{|x|})}), & \text{if } |x| \in (C, \frac{C}{1-q^G(\lambda)}]; \\ (q^G(\lambda), 1), & \text{if } |x| > \frac{C}{1-q^G(\lambda)}, \end{cases} \quad (14)$$

$$\mu^{\text{th}*}(x) = \begin{cases} F^{-1}(1 - P_F^*(x)), & \text{when } x \geq 0; \\ F^{-1}(P_F^*(x)), & \text{when } x < 0, \end{cases}$$

where $q^G(\lambda)$ is the size of the giant component of $\mathcal{G}$, i.e., the largest solution to $q = 1 - e^{-q\lambda}$.

Based on Theorem 1 and 2, the equilibrium spread size and sharing fraction are continuous functions of credibility β in the range $[0, 1]$:

Corollary 1. The steady-state spread size q^* , the sharing fraction P_F^* , and the common sharing threshold $\mu^{\text{th}*}$ are continuous functions of β . Specifically,

$$\lim_{\beta \rightarrow 1} q^*(\beta) = q^*(1).$$

With the focus on threshold sharing strategies (11), the sharing threshold is also continuous at $\beta = 1$. This continuity property will later be used to identify the conditions under which fully credible news does not maximize the cascade size.

3.4. Effects of news magnitude and connectivity on news spread size

Theorem 2 shows that given $\beta = 1$ and the sign of news, the spread size is increasing in news magnitude (cf. (14)). This can be explained by inspecting the sharing decision making in (10). Recall that the persuasive power of news (i.e., $|F(0) - F(\frac{-\beta x}{1-\beta})|$) reaches its full level $\frac{1}{2}$ at $\beta = 1$ regardless of news magnitude. As informed agents are fully convinced and hold their belief means unanimously at the news value, given any q , more extreme news suggests higher expectation about the state, creating a stronger incentive for the informed agents to share the news to persuade their followers. As a result, at the equilibrium the scale of news spread has to be commensurate with the news magnitude. Moreover, for news with moderate magnitude (i.e., $|x| \in (C, \frac{C}{1-q^G(\lambda)})$), the spread size q^* is on the exact level that makes all of informed agents feel indifferent between sharing or not, and the threshold sharing strategy translates to the fraction of sharing P_F^* that sustains the spread size. The above argument for the monotonicity of spread size in news magnitude in fact holds for any level of credibility.

Lemma 4. For any β, λ and F , fixing the sign of news, the equilibrium spread size q^* and the sharing fraction P_F^* are both increasing in news magnitude $|x|$. Moreover, the equilibrium threshold $\mu^{\text{th}*}((x, \beta), F)$ for right-leaning (left-leaning) news is decreasing (increasing) in news magnitude.

Besides the monotone effect of news magnitude, we also demonstrate that higher network connectivity enlarges the scale of diffusion for any news and any credibility.

Lemma 5. Given x, β and F , the equilibrium spread size q^* is increasing in network connectivity λ .

More social connections in the population make the shared news reach to more followers on average, facilitating news diffusion and increasing the scale of spread. However, higher network connectivity in fact yields a smaller fraction of sharing at the equilibrium.

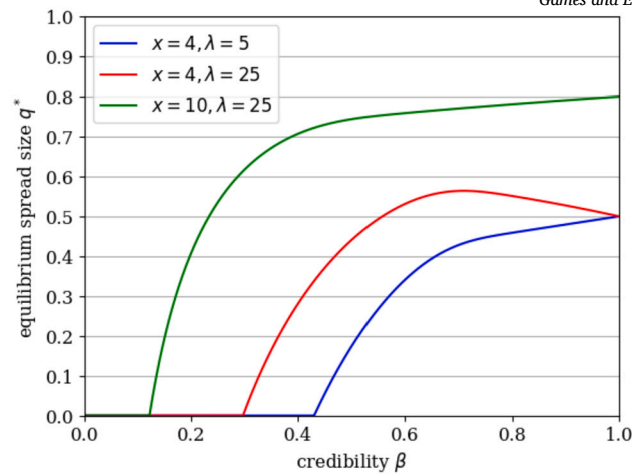


Fig. 1. The equilibrium spread size q^* plotted against credibility β for $C = 2$, $x \in \{4, 10\}$, $\lambda \in \{5, 25\}$ ($q^G(\lambda) \in \{0.993, 0.999\}$), and F following a normal distribution with mean 0 and variance 4, i.e., $F \sim \mathcal{N}(0, 4)$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

It is because, in reaction to the ease of news spread, the informed agents are less motivated to share the news due to the strategic substitutability arising from the cost of sharing.

Though the spread size is increasing both in news magnitude and in network connectivity, we observe that the mechanisms are fundamentally different. For the former, informed agents are incentivized to share extreme news since they have large expectations on the state and the more extreme news can persuade more followers to flip their actions, resulting in a larger fraction of sharing. In contrast, high network connectivity facilitates news spread via more social interactions while discouraging informed agents from sharing. In the next section we will see how such distinct mechanisms affect the relative cascade size of news spread for different levels of credibility.

4. News cascade, scale of spread, and credibility

An intriguing question is how the credibility level may affect the size of news spread in a population. In Theorem 2, we observe that fully credible news ($\beta = 1$) maximizes the size of a news cascade (i.e., the size of the giant component) if news is sufficiently extreme, whereas fully credible news cannot trigger a cascade at all if news is near neutral. For news of moderate magnitude, is the spread size maximized by fully credible news, or can lower credibility induce a larger cascade under certain conditions?

The interplay between credibility and news spread is multifaceted. To illustrate, consider $C = 2$ and agents' perspectives normally distributed with mean 0 and variance 4, i.e., $F \sim \mathcal{N}(0, 4)$. Fig. 1 plots three cases within the moderate news magnitude range ($|x| \in (C, \frac{C}{1-q^G(\lambda)})$), varying with news value and network connectivity.

- The green curve represents the case when $x = 10$, $\lambda = 25$. In this case, news cascades occur for credibility $\beta > 0.1$, with the equilibrium spread size increasing to $1 - \frac{C}{|x|} = 0.8$ at $\beta = 1$.
- The red curve shows that reducing news value to $x = 4$ decreases spread size across all credibility levels at different extent, such that the curve is not monotone in β . The maximum is now achieved at $\beta \approx 0.7$ rather than at $\beta = 1$. This aligns with our understanding that fully credible news may fail to cascade for sufficiently small news values (i.e., $|x| < C$).
- However, when $x = 4$ and connectivity is reduced to an average of 5 followers, as shown by the blue curve, spread size decreases uniformly except at $\beta = 1$, where it remains at 0.5. In this case, $\beta = 1$ maximizes the equilibrium spread size.

These examples illustrate that, for moderate news magnitudes, lower credibility can indeed generate larger cascades than full credible news, indicating its dependence on news value and network connectivity.

To formally analyze these observations, we first derive the comparison of spread size between fully credible and lower credibility news by using Theorem 2 and the continuity of equilibrium spread size at $\beta = 1$ (cf. Corollary 1), as stated in the following proposition:

Proposition 1. Given F and λ , and denote the size of the giant component by $q^G(\lambda)$. Then,

- when news $|x| \leq C$, no informed agents share fully credible news and the cascade size is minimized at $\beta = 1$ with value 0;
- when news $|x| > \frac{C}{1-q^G(\lambda)}$, each informed agent shares fully credible news and the cascade size is maximized at $\beta = 1$ with value $q^G(\lambda)$;

(iii) when news $|x| \in (C, \frac{C}{1-q^G(\lambda)}]$, agents feel indifferent between sharing fully credible news or not sharing it at the equilibrium. In this case, less credible news triggers a larger cascade if and only if the equilibrium threshold for sharing $\mu^{\text{th}*}$ for $\beta = 1$ satisfies²⁶

$$\mu^{\text{th}*}((x, 1), F) > x, \text{ when } x > 0, \quad (15)$$

$$\mu^{\text{th}*}((x, 1), F) < x, \text{ when } x < 0.$$

where

$$\mu^{\text{th}*}((x, 1), F) = \begin{cases} F^{-1}\left(1 - \frac{-\ln(\frac{C}{|x|})}{\lambda(1 - \frac{C}{|x|})}\right), & \text{when } x > 0; \\ F^{-1}\left(\frac{-\ln(\frac{C}{|x|})}{\lambda(1 - \frac{C}{|x|})}\right), & \text{when } x < 0. \end{cases}$$

For the ease of exposition, we discuss the implications of Proposition 1 and the following results for right-leaning news $x > 0$; the discussion for left-leaning news $x < 0$ is similar. We focus on the range of news of moderate magnitude (cf. case (iii)). Indeed, the high connectivity that is usually observed on empirical social networks (cf. footnote 8) suggests that the giant component size $q^G(\lambda)$ is nearly 1.²⁷ This implies that even when the cost of sharing is small, $\frac{C}{1-q^G(\lambda)}$ can still be significantly large and the interval $(C, \frac{C}{1-q^G(\lambda)})$ can cover a wide range of news magnitude of interest.

The rationale for the condition (15) in case (iii) is as follows. Fix the spread size $q^* = 1 - \frac{C}{|x|}$ that corresponds to the case $\beta = 1$ while reducing credibility β from 1. According to the decision rule of sharing (10), the effective persuasive power of news remains nearly the same as $\frac{1}{2}$, suggesting that informed agents with perspectives greater than x strictly prefer to share (β less than 1). When the condition $\mu^{\text{th}*}((x, 1), F) > x$ holds, more informed agents are incentivized to share the less credible news, suggesting that spread size q^* can not be sustained and the scale of news increases.

We utilize Proposition 1 to identify the range of news values where fully credible news does not maximize spread size. Lemma 4 states that the equilibrium threshold $\mu^{\text{th}*}((x, \beta), F)$ is decreasing (resp. increasing) in the magnitude of right-leaning news (resp. left-leaning news). Consequently, condition (15) suggests that lower credibility news generates larger cascades for news that is not sufficiently extreme, as observed in Fig. 1 when news magnitude decreases from 10 to 4. This finding is formalized in the following proposition:

Proposition 2. Given λ , F , and the sign of news, the spread of lower credibility news is larger than fully credible news for news with sufficiently small magnitude. Specifically, there exist two unique thresholds $\underline{x}(\lambda, F) < 0 < \bar{x}(\lambda, F)$ such that the spread of lower credibility news is larger than fully credible news if and only if $x \in (\underline{x}, \bar{x})$.

When news is small or almost neutral, high credibility will concentrate the beliefs of informed agents to be around small positive values. As agents find low expected values for the state, only those agents with quite extreme perspectives (who hold relatively extreme updated belief mean after seeing the news) share the news, yielding a large threshold $\mu^{\text{th}*}((x, \beta), F)$. On the other hand, when the credibility is reduced, informed agents are less convinced and their updated expectations about the state are more dispersed compared to observing fully credible news. Despite the compromised persuasiveness of news due to lower credibility that discourages agents from sharing, there are more informed agents who still perceive high expected values for the state and have strong incentives to share, propelling the spread of news with lower credibility. This result mirrors the earlier work of Acemoglu et al. (2024), which, considering news value at a continuous scale, suggested that reduced extremism of news facilitates the virality of misinformation.

We now examine how network connectivity affects the relative spread size between fully credible and lower credibility news. In Fig. 1 increased connectivity appears to favor wider spread of lower credibility news. Indeed, for news with $|x| > C$, there exists a threshold $\lambda^{\text{th}}(x, F)$ such that lower credibility news trigger larger cascades if and only if $\lambda > \lambda^{\text{th}}(x, F)$. To see this, we note from Theorem 2 that given news $x > C$ ($x < -C$) and $\beta = 1$, the equilibrium sharing fraction P_F^* is decreasing in λ , while the equilibrium threshold $\mu^{\text{th}*}$ is increasing (decreasing) in λ . Specifically, for small connectivity λ , we have $\frac{C}{1-q^G(\lambda)} \leq |x|$ and $P_F^* = 1$. This fraction begins to decrease when λ is large enough that $\frac{C}{1-q^G(\lambda)} > |x|$. Then condition (15) holds if λ exceeds a threshold, which can be derived by measuring the fraction of agents who share news based on $\mu^{\text{th}*}((x, 1), F)$ and x in condition (15), through an indifference equation:

²⁶ Hsu et al. (2021) assumed F to be a normal distribution with zero mean. Under this specific setting the corresponding version of condition (15) is also necessary for $\beta = 1$ not to be the global maximizer of the cascade size. However, the authors did not further quantify the effects of x , λ and variance of the distribution on this sufficient and necessary condition.

²⁷ Using the empirical average degree $\lambda = 35.2$ for Poisson network, the giant component is computationally 1.

$$\underbrace{\frac{-\ln \frac{C}{|x|}}{\lambda^{\text{th}}(1 - \frac{C}{|x|})}}_{=P_F^*(x)} = \begin{cases} 1 - F(x), & \text{when } x > 0; \\ F(x), & \text{when } x < 0. \end{cases} \quad (16)$$

The next corollary formalizes this observation:

Proposition 3 (Connectivity limit). *Given F , for news x with magnitude $|x| > C$, lower credibility news can trigger a larger cascade than fully credible news when the network connectivity is sufficiently high. Specifically there exists a unique value $\lambda^{\text{th}}(x, F)$ such that lower credibility news results in a cascade with larger size than fully credible news if and only if $\lambda > \lambda^{\text{th}}(x, F)$ where*

$$\lambda^{\text{th}}(x, F) = \begin{cases} \frac{-\ln \frac{C}{|x|}}{(1 - \frac{C}{|x|})(1 - F(x))}, & \text{when } x > 0; \\ \frac{-\ln \frac{C}{|x|}}{(1 - \frac{C}{|x|})F(x)}, & \text{when } x < 0. \end{cases}$$

Moreover, given the sign of news, $\lambda^{\text{th}}(x, F)$ is increasing in news magnitude $|x|$. We refer to the threshold λ^{th} as network connectivity limit.

Acknowledging the aiding effect of network connectivity on news diffusion (cf. Lemma 5), informed agents are discouraged from sharing the news due to the strategic substitutability; their updated expected values about the state have to be more extreme so that they will share, which suggests an increased threshold for the equilibrium sharing strategy.

However, the extent of such increment in the threshold varies significantly for different levels of news credibility: The indifference equation for the sharing threshold (cf. Theorem 1) shows that a small increment in spread size q (thanks to larger connectivity) will be translated to a significant increase in μ^{th} when credibility level β is high. Indeed, in this case the updated means barely allow for subjective perspectives, the threshold on perspective has to be much more extreme to result in large enough updated belief means that motivate sharing. In contrast, when observing news with lower credibility than full level, agents with large right-leaning perspectives still hold large updated belief means and would like to share the news. Consequently, on a densely-connected network, lower credibility news triggers a cascade that outsizes the spread of fully credible news.

Proposition 3 suggests that the connectivity limit $\lambda^{\text{th}}(x, F)$ serves as a natural index for the level of susceptibility of a population to wider spread of lower credibility news than credible news: The network connectivity can not surpass the threshold λ^{th} or lower credibility news can generate a larger cascade than fully credible news. This property enables us to evaluate the impact of perspective CDF F on the virality of lower credibility news by examining its influence on connectivity limit. In the next section, we study how the distribution of perspectives in a polarized population may facilitate spread of news with lower credibility.

5. News cascade in populations with polarized perspectives

In this section, we study the societies comprising two groups of agents who exhibit polarization in their perspectives on the state. We investigate how the extent of polarization and the diversity of perspectives within each group can affect the spread of news for different levels of credibility.

Recognizing bimodality as a key attribute of polarized distributions (Fiorina et al., 2005; Lelkes, 2016; Bramson et al., 2017), we use a parametric family of distributions that captures this bimodality while providing degrees of freedom for varying shapes within these distributions. Formally, we assume that each agent i is equally likely to orient her perspective toward either left or right side of perspective spectrum, which is her private information and denoted as $e_i \in \{-1, +1\}$. We refer to the group of agents with orientation -1 ($+1$) as left-wing (right-wing). Agent i 's perspective is then independently drawn from a distribution that conditional on her orientation e_i :

$$\mu_i | e_i \sim \begin{cases} \frac{1}{\sigma_\mu} g\left(\frac{\mu + \bar{\mu}}{\sigma_\mu}\right), & \text{when } e_i = -1; \\ \frac{1}{\sigma_\mu} g\left(\frac{\mu - \bar{\mu}}{\sigma_\mu}\right), & \text{when } e_i = +1, \end{cases} \quad (17)$$

where g is a continuous density function, $\bar{\mu} \geq 0$ measures the *polarization* between the two groups, and $\sigma_\mu > 0$ represents the diversity of perspectives within each group. We will also refer to σ_μ as *in-group diversity*. Similar to Assumption 2, we impose regularity conditions on density function g :

Assumption 4. The density function g is continuously-differentiable, log-concave, and symmetric around zero, with full support on the real line, with mean and variance. We use G to represent the CDF of g .

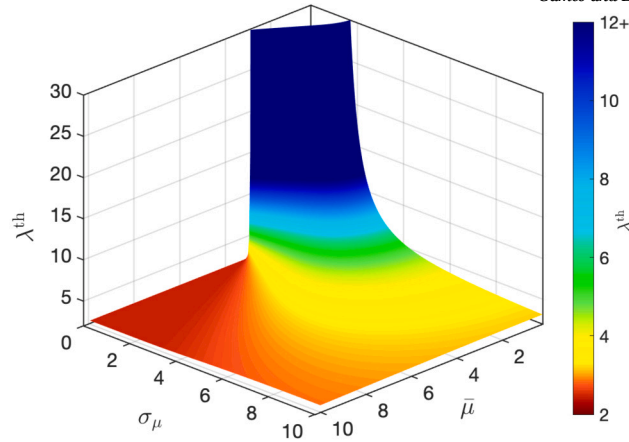


Fig. 2. The network connectivity limit $\lambda^{\text{th}}(x, \bar{\mu}, \sigma_\mu, G)$ as a function of polarization $\bar{\mu}$ and in-group diversity σ_μ when $C = 2, x = 3$, and G is the CDF for standard normal distribution. The colormap indicates the value of connectivity limit.

The CDF F of perspectives given polarization $\bar{\mu}$ and in-group diversity σ_μ can be expressed as

$$F(\mu) = \frac{1}{2}G\left(\frac{\mu + \bar{\mu}}{\sigma_\mu}\right) + \frac{1}{2}G\left(\frac{\mu - \bar{\mu}}{\sigma_\mu}\right). \quad (18)$$

Note that $F(0)$ is a constant $\frac{1}{2}$ and also $F(-\mu) = 1 - F(\mu)$ for any μ . The distributions (17) and (18) are common knowledge.

From Proposition 1, we can see that lower credibility ($\beta < 1$) can trigger a larger cascade than fully credible news ($\beta = 1$) only if the news has sufficiently small magnitude ($|x| < C$) or the network is sufficiently connected ($\lambda > \lambda^{\text{th}}(x, F)$). Below we investigate the effects of the parameters $\bar{\mu}, \sigma_\mu, G$ on the connectivity limit, using the more explicit notation $\lambda^{\text{th}}(x, \bar{\mu}, \sigma_\mu, G)$. We rewrite the connectivity limit given in (16), for any news x , as

$$\lambda^{\text{th}}(x, \bar{\mu}, \sigma_\mu, G) = \frac{-\ln \frac{C}{|x|}}{\left(1 - \frac{C}{|x|}\right)F(-|x|)} = \frac{-\ln \frac{C}{|x|}}{\left(1 - \frac{C}{|x|}\right)\left(\frac{1}{2}G\left(\frac{-|x| + \bar{\mu}}{\sigma_\mu}\right) + \frac{1}{2}G\left(\frac{-|x| - \bar{\mu}}{\sigma_\mu}\right)\right)}. \quad (19)$$

The first result concerns the impact of polarization on the network connectivity limit:

Proposition 4 (Effect of polarization). For any density g , in-group diversity σ_μ , and any news x with magnitude $|x| > C$, increased polarization decreases network connectivity limit, i.e., λ^{th} is decreasing in $\bar{\mu}$. Moreover, $\lambda^{\text{th}}(x, \bar{\mu} = 0, \sigma_\mu, G) = \frac{-\ln \frac{C}{|x|}}{\left(1 - \frac{C}{|x|}\right)G\left(-\frac{|x|}{\sigma_\mu}\right)}$.

Stronger polarization in a population leads to a larger fraction of extremists, who, as we have already argued, assume a higher value for the persuasiveness effect of the news aligned with their large belief mean when $\beta < 1$. Though stronger polarization reduces the fraction of the extremists who have the opposite ideological leaning from their group (e.g., right-leaning agents in left-wing group), such extremists only account for a small fraction (on the tail of the perspective distribution of left-wing group). As a result, the fraction of extremists overall increases, facilitating the sharing of lower credibility news. The news with lower credibility can hence trigger a larger cascade than fully credible news even on the networks with smaller connectivity, suggesting a decreased connectivity limit.

Fig. 2 illustrates the effect of polarization $\bar{\mu}$ and in-group diversity σ_μ on network connectivity limit λ^{th} when function g is the standard normal distribution. Given any level of in-group diversity, we can see that the network connectivity limit decreases as the polarization is increased.

On the other hand, the effect of increased in-group diversity on the network connectivity limit is not necessarily monotone, as stated in the following proposition:

Proposition 5 (Effect of in-group diversity). For any density g (and corresponding G) and any news x with magnitude $|x| > C$,

- when polarization is weak, increased in-group diversity decreases the network connectivity limit. Specifically, when $\bar{\mu} \leq |x|$, λ^{th} is decreasing in σ_μ and moreover, $\lim_{\sigma_\mu \rightarrow 0} \lambda^{\text{th}}(x, \bar{\mu}, \sigma_\mu, G) = +\infty$;
- when polarization is strong, increased in-group diversity increases the network connectivity limit only when the in-group diversity is small. Formally, when $\bar{\mu} > |x|$, λ^{th} is increasing in σ_μ up to some $\hat{\sigma}(x, \bar{\mu}, G) > 0$ and then decreasing. Moreover, $\lim_{\sigma_\mu \rightarrow 0} \lambda^{\text{th}}(x, \bar{\mu}, \sigma_\mu, G) = \frac{-2 \ln \frac{C}{|x|}}{1 - \frac{C}{|x|}}$.

In a population with weak polarization and low in-group diversity, the agents' perspectives are highly concentrated around the neutrality with a small fraction of extremists. Increasing the in-group diversity from a low level instead results in more agents with extreme perspectives for both ideological leanings, lowering the network connectivity limit. Fig. 2 exemplifies that given some polarization level $\bar{\mu} \leq x$ the network connectivity limit approaches the infinity when the diversity is nearly zero and that the connectivity limit is decreasing in diversity.

On the other hand, in a population with strong polarization, increased in-group diversity has a non-monotone effect on the connectivity limit. In this case, there is a large fraction of extremists and enlarged diversity affects the fraction of extremists in two ways: (i) decreases the number of the extremists whose perspectives align with their group's orientation, while (ii) *increases* the number of the extremists whose perspectives are opposing to their group's orientation and are on the tail of the perspective distribution of the group.

When there is low in-group diversity in the population, the decrease from (i) outsizes the increase from (ii), reducing the fraction of extremists and raising up the connectivity limit. For example, with fixed $\bar{\mu} = 5$ in Fig. 2, λ^{th} increases from 2.5 to around 4 as σ_{μ} is increased from 0 to 5. However, due to the log-concavity of the perspective distribution (i.e., g), the relative magnitude of effect (i) to (ii) is decreasing in diversity and consequently the network connectivity limit starts to fall when the diversity becomes sufficiently large. We can see in Fig. 2 that given $\bar{\mu} = 5$, λ^{th} is decreasing for $\sigma_{\mu} > 5$.

6. Conclusions

In this paper, we studied strategic news-sharing decisions on social networks when individuals' main motive for sharing is to persuade their followers. We postulated that agents have an incentive to persuade their followers to action according to their beliefs by sharing persuasive news and not disclosing news that may move them to action in the opposite direction. Sharing decisions feature strategic substitutes since an agent may find sharing not beneficial when it is likely that her followers may receive the news from others. We characterized the Bayesian Nash equilibrium of the news-sharing game, identifying individual decision rule of sharing and the resulting endogenous news cascade. We further delineated the sufficient and necessary conditions under which news with lower credibility can trigger a larger spread than fully credible news. As our main result, we showed that low credibility news can result in a larger cascade than fully credible news when the network is highly connected. Moreover, we investigated news spread in polarized populations. We found that strong polarization relaxes the connectivity limit necessary for lower credibility news to generate a larger cascade. We also illustrated that increased diversity of beliefs within each ideological group does not necessarily reduce the spread of news with lower credibility: When the polarization is weak or when both groups have quite diverse perspectives, raising up the level of in-group diversity can instead prompt more sharing of lower credibility news.

Our model centered on informative news with known credibility, both measured on a continuous scale. Future research should consider the complexities of fake news, which offers no genuine informational value. Fake news cannot be distinguished from real news without verification and individuals can only speculate on news veracity based on news value and their prior beliefs on news production process. Building on the findings by Hsu et al. (2022), who investigated Bayesian agents' reactions to unverified news, we anticipate such an extension to our model would maintain the equilibria featuring threshold sharing strategies. However, it could introduce nuanced effects on news spread, yielding new insights into the interplay of news value, population characteristics, and network structure. Furthermore, incorporating costly verification options would also provide a valuable extension.

Declaration of competing interest

This is to certify, on behalf of the authors, that there are no interests to declare for above paper.

Appendix A

Proof of Lemma 2. Given F, δ , denote by Q the probability that a follower has received the news at the steady state from other agents according to the strategy profile s ; we reason similarly as how the equation for q is derived. However, we need to adjust two elements: (i) the out-degree distribution of a follower should be adjusted by the well-known *friendship paradox*; (ii) a follower with out-degree d has not received it from any of her "other" followees, conditioned on the event that the decision maker does not share the news.

For (i), denote by $P_f^{\text{out}}(d)$ the probability that a follower has out-degree d ($d \geq 1$ since the follower follows the agent). By our model, this quantity is proportional to $d P^{\text{out}}(d)$. We derive the proper distribution by normalization:

$$P_f^{\text{out}}(d) = \frac{d P^{\text{out}}(d)}{\sum_{d=0}^{+\infty} d P^{\text{out}}(d)} = \frac{d P^{\text{out}}(d)}{\mathbb{E}[d]}.$$

For (ii), conditional on the follower having out-degree d , the probability that she does not hear from any other followees at the steady state (i.e., the followees do not receive any news or they do not share it) is $(1 - q P_F(s((x, \beta)))^{d-1}$. Using the law of total probability, we can deduce the following equation that quantifies $1 - Q$ for the out-degree distribution $P^{\text{out}}(\cdot)$:

$$1 - Q = (1 - \delta) \sum_{d=1}^{\infty} P_f^{\text{out}}(d) (1 - q P_F(s((x, \beta)))^{d-1}.$$

When the out-degree distribution follows a Poisson distribution with mean λ , i.e., $P^{\text{out}}(\cdot) \sim \text{Poisson}(\lambda)$, we can derive that

$$P_f^{\text{out}}(d) = \frac{d \frac{1}{d!} \lambda^d e^{-\lambda}}{\lambda} = \frac{\lambda^{d-1} e^{-\lambda}}{(d-1)!},$$

and that

$$1 - Q = (1 - \delta) \sum_{d=1}^{\infty} \frac{\lambda^{d-1} e^{-\lambda}}{(d-1)!} (1 - q P_F(s((x, \beta)))^{d-1} = (1 - \delta) \sum_{d=0}^{\infty} P^{\text{out}}(d) (1 - q P_F(s((x, \beta)))^d.$$

We obtain that the probability Q equals the δ -spread size q . $\square$

Proof of Lemma 3. According to (2), we obtain

$$\begin{aligned} & \mathbb{E}_i[a_k | k \in N_i^{\text{in}}, z_k = (x, \beta)] \\ &= \mathbb{P}_{\mu_k \sim F}[a_k(\mu_k, (x, \beta)) = +1] - \mathbb{P}_{\mu_k \sim F}[a_k(\mu_k, (x, \beta)) = -1] \\ &= 1 - 2F\left(\frac{-\beta x}{1 - \beta}\right), \end{aligned}$$

and similarly,

$$\mathbb{E}_i[a_k | k \in N_i^{\text{in}}, z_k = \emptyset] = 1 - 2F(0).$$

Plug them into (8) and the sharing rule (10) is the immediate result. $\square$

Proof of Theorem 1. WLOG, we discuss the case $x \geq 0$. The equilibrium when $\delta \rightarrow 0$ should satisfy the following system of equations:

$$\begin{aligned} q^* &= 1 - e^{-q^* \lambda P_F(s^*)}, \\ P_F(s^*) &= 1 - F\left(\frac{\frac{C}{2(1-q^*)F^{\text{flip}}(x, \beta)} - \beta x}{1 - \beta}\right) \end{aligned}$$

To prove the existence and uniqueness of an equilibrium, we define a continuous function $h(q)$ for $q \in [0, 1]$:

$$\begin{aligned} h(q) &\triangleq 1 - e^{-q \lambda P_F(s)} - q, \\ P_F(s) &= 1 - F\left(\frac{\frac{C}{2(1-q)F^{\text{flip}}(x, \beta)} - \beta x}{1 - \beta}\right) \end{aligned} \quad (20)$$

By the continuity and $h(0) = 0$ and $h(1) = -1$ (because $P_F(s) = 0$ for $q = 1$), there must exist at least one such solution to (20), proving the existence of an equilibrium. Though $q = 0$ is always a solution, the zero spread size may not be an equilibrium for infinitesimal seeding size δ when there are other positive solutions (see Jackson (2008) for details). We now identify the condition under which there are other positive solutions to $h(q)$, i.e., a news cascade emerges, and then show the uniqueness of the equilibrium.

Emergence of a news cascade

Take derivative (20) of w.r.t. q ,

$$h'(q) = \lambda e^{-\lambda q P_F(s)} \left(P_F(s) + q \frac{dP_F(s)}{dq} \right) - 1.$$

Note that for $\beta < 1$, P_F is differentiable in the range $[0, 1]$ (for $x = 0$ or $\beta = 0$, $P_F(s)$ is the constant zero). As $h(0) = 0$, $h(1) = -1$, if $h'(0) = -1 + \lambda P_F(s)|_{q=0} > 0$, due to the continuity, some positive solutions must exist.

We now show that $h'(0) > 0$ is also necessary. Noting that the derivative $\frac{dP_F(s)}{dq}$ must be non-positive due to the strategic substitutability and $P_F(s)$ is maximized at $q = 0$. Consequently, we can find that when $h'(0) \leq 0$, then $h'(q) < 0$ for any $q > 0$, indicating that $h(q)$ is strictly decreasing and $q = 0$ is the only root.

Uniqueness of the equilibrium

We have shown that if $\lambda P_F(s)|_{q=0} \leq 1$, $q^* = 0$ is the only solution and $P_F^*, \mu^{\text{th}*}$ are in turn determined, resulting in a unique equilibrium. If $\lambda P_F(s)|_{q=0} > 1$, there are other positive solutions and $q^* \neq 0$ (Jackson, 2008); for this case, there is only one positive solution and the equilibrium is therefore unique. To see this, suppose there are two positive solutions $q^1 > q^2 > 0$. Then we have $P_F(s)|_{q=q^1} < P_F(s)|_{q=q^2}$ due to strategic substitutability. However, the equation $q = 1 - e^{-q \lambda P_F(s)}$, or $P_F(s) = \frac{-\ln(1-q)}{\lambda q}$, dictates that the solutions should satisfy $P_F(s)|_{q=q^1} > P_F(s)|_{q=q^2}$. Contradiction! As a result, only one positive solution for q exists, with $P_F^*, \mu^{\text{th}*}$ in turn determined, and the equilibrium is unique. $\square$

Proof of Theorem 2. For convenience, define $\tilde{q}(x) \triangleq 1 - \frac{C}{|x|}$. Given λ, F , for $\beta = 1$, according to Lemma 3, we can see that $P_F(s(x, 1)) = 1$ for any $q < \tilde{q}(x)$ while $P_F(s(x, 1)) = 0$ for any $q > \tilde{q}(x)$. We keep the argument news x and β only as a shorthand.

Note that the cascade size is bounded above by the size of giant component $q^G(\lambda)$ when every agent shares news ($P_F \equiv 1$). We have three cases depending on the magnitude of x :

- 1) $|x| \in [0, C]$: $\tilde{q}(x, 1) \leq 0$ and then $P_F(x, 1) = 0$ for any $q \in (0, 1] \Rightarrow q^*(x, 1) = 0, P_F^*(x, 1) = 0$.
- 2) $|x| \in (C, \frac{C}{1-q^G(\lambda)})$: In this case $\tilde{q}(x) \in [0, q^G(\lambda)]$. Observe that $h(q) > 0$ for $q < \tilde{q}(x)$ and $h(q) < 0$ for $q > \tilde{q}(x)$. Let $P_F(x, 1)|_{q=\tilde{q}(x)} = \frac{-\ln(1-\tilde{q}(x))}{\lambda\tilde{q}(x)}$, and it can be shown that $P_F(x, 1) \in [0, 1]$ and $h(\tilde{q}(x)) = 0$. Therefore, the equilibrium is $q^*(x, 1) = \tilde{q}(x)$ and $P_F^*(x, 1) = \frac{-\ln(1-\tilde{q}(x))}{\lambda\tilde{q}(x)}$.
- 3) $|x| > \frac{C}{1-q^G(\lambda)}$: $\tilde{q}(x) > q^G(\lambda)$ and then $P_F(x) = 1$ for any $q \leq q^G(\lambda)$. It is obvious that $q^*(x, 1) = q^G(\lambda), P_F^*(x, 1) = 1$.

The equilibrium sharing threshold $\mu^{\text{th}*}(x, 1)$ is obtained by applying inverse of CDF F^{-1} to the sharing fraction $P_F^*(x, 1)$. The existence of an equilibrium with is then obvious. $\square$

Proof of Corollary 1. When $\beta < 1$, it is easy to see that for any news x , $q^*(x, \beta)$, $P_F^*(x, \beta)$ and $\mu^{\text{th}*}(x, \beta)$ are continuous in $\beta \in [0, 1)$. We now show that they are continuous at $\beta = 1$, i.e., their limits when $\beta \rightarrow 1$ are equal to their corresponding values at $\beta = 1$.

- 1) $|x| \leq [0, C]$: If $\lim_{\beta \rightarrow 1} q^*(x, \beta) > 0$, then for $x \geq 0$

$$\lim_{\beta \rightarrow 1} \mu^{\text{th}*}(x, \beta) = \lim_{\beta \rightarrow 1} \frac{\frac{C}{2(1-q^*(x, \beta))F(\frac{-\beta x}{1-\beta})} - \beta x}{1 - \beta} = \lim_{\beta \rightarrow 1} \frac{\frac{C}{1-q^*} - x}{1 - \beta} = +\infty,$$

and hence $\lim_{\beta \rightarrow 1} P_F^*(x, \beta) = 0$, resulting in $\lim_{\beta \rightarrow 1} q^*(x, \beta) = 0$. Then obviously $\lim_{\beta \rightarrow 1} q^*(x, \beta) = 0$ and $\lim_{\beta \rightarrow 1} P_F^*(x, \beta) = 0$. The reasoning for $x < 0$ is similar.

- 2) $|x| \in (C, \frac{C}{1-q^G(\lambda)})$: If $\lim_{\beta \rightarrow 1} q^*(x, \beta) < \tilde{q}(x)$, then $\lim_{\beta \rightarrow 1} P_F^*(x, \beta) = 1$, resulting in $\lim_{\beta \rightarrow 1} q^*(x, \beta) = q^G(\lambda) \geq 1 - \frac{C}{|x|}$. Contradiction. Similarly $\lim_{\beta \rightarrow 1} q^*(x, \beta) > \tilde{q}(x)$ can not hold as well. Therefore $\lim_{\beta \rightarrow 1} q^*(x, \beta) = \tilde{q}(x)$. Moreover, $P_F^*(x, 1) = \frac{-\ln(1-\tilde{q}(x))}{\lambda\tilde{q}(x)}$.
- 3) $|x| > \frac{C}{1-q^G(\lambda)}$: Since q^* is bounded above by $q^G(\lambda)$, we hence obtain $\lim_{\beta \rightarrow 1} P_F^*(x, \beta) = 1$, which sustains that $\lim_{\beta \rightarrow 1} q^*(x, \beta) = q^G(\lambda)$.

Therefore, with the results from Theorem 2, $q^*(x, \beta)$ and $P_F^*(x, \beta)$ are both continuous at $\beta = 1$. As $\mu^{\text{th}*}(x, \beta)$ is a continuous function of $P_F^*(x, \beta)$ (inverse CDF F^{-1}), it is also continuous at $\beta = 1$. $\square$

Proof of Lemma 4. It is self-evident for the case when $\beta = 1$ by Theorem 2. For $\beta < 1$, we observe that, given news sign, if for any $q \in [0, 1]$, the corresponding value of function $1 - e^{-q\lambda P_F(s((x, \beta), F))}$ is increased in $|x|$, the solution q^* (and corresponding fraction of sharing P_F^*) is increased as well. Therefore, it suffices to show that for any q , P_F is increasing in news magnitude $|x|$; this can be

easily proved by noting the monotonicity of the sharing threshold $\mu^{\text{th}}((x, \beta), F) = \frac{\frac{C}{2(1-q^*(x, \beta))F(\frac{-\beta x}{1-\beta})} - \beta x}{1 - \beta}$ in news magnitude. $\square$

Proof of Lemma 5. According to Theorem 2, given $\beta = 1$, when $|x| \leq C$ the equilibrium spread size is zero whereas when $|x| > C$ it is $\min\{1 - \frac{C}{|x|}, q^G(\lambda)\}$; in either case the equilibrium spread size is increasing in λ . For $\beta < 1$, the argument is similar to the above proof: Since all else equal the function value of $1 - e^{-q\lambda P_F(s((x, \beta), F))}$ is increasing in λ , the result is then obvious. $\square$

Proof of Proposition 1. Case (i) and (ii) are direct results of Theorem 2. We now prove the necessary and sufficient condition for $\beta = 1$ to be a local minimizer when $|x| \in (C, \frac{C}{1-q^G(\lambda)})$. We compute the derivative of q^* w.r.t. β when $\beta \rightarrow 1$, i.e., $\lim_{\beta \rightarrow 1} \frac{\partial q^*}{\partial \beta}$. In the following we only compute the case for $x > 0$ and for $x < 0$ the approach is similar. To evaluate $\lim_{\beta \rightarrow 1} \frac{\partial q^*}{\partial \beta}$, we use the continuity of $\mu^{\text{th}*}$ at $\beta = 1$.

$$\mu^{\text{th}*}((x, 1), F) = \lim_{\beta \rightarrow 1} \mu^{\text{th}*}((x, \beta), F) = \lim_{\beta \rightarrow 1} \frac{\frac{C}{2(1-q^*(x, \beta))F^{\text{flip}}(x, \beta)} - \beta x}{1 - \beta}.$$

Applying l'hopital rule to the last term and using the fact that $\lim_{|\mu| \rightarrow \infty} \mu^2 f(\mu) = 0$ (as the mean and variance are both finite), we derive

$$\lim_{\beta \rightarrow 1} \frac{\frac{C}{2(1-q^*)F^{\text{flip}}(x, \beta)} - \beta x}{1 - \beta} = \lim_{\beta \rightarrow 1} \frac{-C \left(-\frac{dq^*}{d\beta} F^{\text{flip}}(x, \beta) + (1-q^*)f\left(\frac{-\beta x}{1-\beta}\right) \frac{x}{(1-\beta)^2} \right)}{2((1-q^*)F^{\text{flip}}(x, \beta))^2} = \frac{-x^2 \lim_{\beta \rightarrow 1} \frac{dq^*}{d\beta}}{C} + x.$$

As a result, when $x > 0$, $\lim_{\beta \rightarrow 1} \frac{dq^*}{d\beta} < 0$ iff $\mu^{\text{th}*}((x, 1), F) > x$. Similarly, repeating the approach, one can show that when $x < 0$, $\lim_{\beta \rightarrow 1} \frac{dq^*}{d\beta} < 0$ iff $\mu^{\text{th}*}((x, 1), F) < x$. The proof is completed. $\square$

Proof of Proposition 2. WLOG, we prove for positive news; we aim to show the existence of $\bar{x}(\lambda, F)$.

According to Proposition 1, for $\beta = 1$, when $x \leq C$, no agents share the news whereas when $x > \frac{C}{1-qG(\lambda)}$, every agent shares the news. It also means that $\mu^{\text{th}*}(\{x = C, 1\}, F) = +\infty$ and $\mu^{\text{th}*}(\{x = \frac{C}{1-qG(\lambda)}, 1\}, F) = -\infty$. Therefore, in the range $(C, \frac{C}{1-qG(\lambda)})$, the equilibrium sharing threshold $\mu^{\text{th}*}((x, 1), F)$ is decreasing in x , from $+\infty$ to $-\infty$ (cf. Lemma 4). We can then easily see that there exists unique $\bar{x}(\lambda, F) > C$ such that $\mu^{\text{th}*}(\{x, 1\}, F) > x$ iff $x < \bar{x}(\lambda, F)$. $\square$

Proof of Proposition 4. Observing that in (19), $\bar{\mu}$ affects λ^{th} through function $F(-|x|)$, we investigate their effects on $F(-|x|)$.

When $\bar{\mu} = 0$, $\lambda^{\text{th}}(x, 0, \sigma_\mu, G) = \frac{-\ln \frac{C}{|x|}}{(1-\frac{C}{|x|})G(-\frac{|x|}{\sigma_\mu})}$. The derivative of $F(-|x|)$ w.r.t. $\bar{\mu}$ is

$$\frac{dF(-|x|)}{d\bar{\mu}} = \frac{1}{2\sigma_\mu} \left[g\left(\frac{-|x| + \bar{\mu}}{\sigma_\mu}\right) - g\left(\frac{-|x| - \bar{\mu}}{\sigma_\mu}\right) \right] = \frac{1}{2\sigma_\mu} \left[g\left(\frac{\bar{\mu} - |x|}{\sigma_\mu}\right) - g\left(\frac{\bar{\mu} + |x|}{\sigma_\mu}\right) \right] > 0,$$

since g is even and decreasing in $[0, +\infty)$ and $|\bar{\mu} + |x|| > |\bar{\mu} - |x||$. Therefore $F(-|x|)$ is strictly increasing in $\bar{\mu}$ and then $\lambda^{\text{th}}(x, \bar{\mu}, \sigma_\mu, G)$ is decreasing in $\bar{\mu}$. $\square$

Proof of Proposition 5. We can similarly derive the effect of σ_μ on λ^{th} by examining its effect on $F(-|x|)$:

$$\begin{aligned} \frac{dF(-|x|)}{d\sigma_\mu} &= \frac{1}{2\sigma_\mu} \left[g\left(\frac{-|x| + \bar{\mu}}{\sigma_\mu}\right) \left(\frac{x - \bar{\mu}}{\sigma_\mu}\right) + g\left(\frac{-|x| - \bar{\mu}}{\sigma_\mu}\right) \left(\frac{x + \bar{\mu}}{\sigma_\mu}\right) \right], \\ &= \frac{1}{2\sigma_\mu} \left[g\left(\frac{\bar{\mu} - |x|}{\sigma_\mu}\right) \left(\frac{x - \bar{\mu}}{\sigma_\mu}\right) + g\left(\frac{\bar{\mu} + |x|}{\sigma_\mu}\right) \left(\frac{\bar{\mu} + x}{\sigma_\mu}\right) \right], \end{aligned}$$

which is positive iff

$$\frac{g\left(\frac{\bar{\mu} + |x|}{\sigma_\mu}\right)}{g\left(\frac{\bar{\mu} - |x|}{\sigma_\mu}\right)} > \frac{\bar{\mu} - |x|}{\bar{\mu} + |x|}. \quad (21)$$

Note that for $\bar{\mu} \leq |x|$, (21) holds for any σ_μ , implying that $F(-|x|)$ is increasing in σ_μ . When $\bar{\mu} > |x|$, we first show that the LHS of (21) is increasing in σ_μ . Take the derivative, we find that

$$\frac{d}{d\sigma_\mu} \frac{g\left(\frac{\bar{\mu} + |x|}{\sigma_\mu}\right)}{g\left(\frac{\bar{\mu} - |x|}{\sigma_\mu}\right)} = - \underbrace{\frac{g\left(\frac{\bar{\mu} + |x|}{\sigma_\mu}\right)}{g\left(\frac{\bar{\mu} - |x|}{\sigma_\mu}\right)} \frac{(\bar{\mu} + |x|)}{\sigma_\mu^2} - \frac{g'\left(\frac{\bar{\mu} - |x|}{\sigma_\mu}\right)}{g\left(\frac{\bar{\mu} - |x|}{\sigma_\mu}\right)} \frac{(\bar{\mu} - |x|)}{\sigma_\mu^2}}_{(*)}.$$

Since $\frac{g'(y)}{g(y)}$ is decreasing due to log-concavity of g and $\frac{g'(y)}{g(y)} < 0$ for $y > 0$, the part $(*)$ is negative and consequently LHS of (21) is increasing in σ_μ . Observing that LHS of (21) approaches to 0 when $\sigma_\mu \rightarrow 0$ while to 1 when $\sigma_\mu \rightarrow +\infty$, the monotonicity then suggests that there exists some value $\hat{\sigma}(x, \mu, G) > 0$ such that $F(-|x|)$ is decreasing in σ_μ iff $\sigma_\mu \leq \hat{\sigma}(x, \mu, G)$.

For the two cases discussed above:

- When $\bar{\mu} \leq |x|$, we have $\lim_{\sigma_\mu \rightarrow 0} F(-|x|) = 0$. Hence, $\lim_{\sigma_\mu \rightarrow 0} \lambda^{\text{th}}(x, \bar{\mu}, \sigma_\mu, G) = +\infty$ and $\lambda^{\text{th}}(x, \bar{\mu}, \sigma_\mu, G)$ is decreasing in σ_μ .
- When $\bar{\mu} > |x|$, we have $\lim_{\sigma_\mu \rightarrow 0} F(-|x|) = \frac{1}{2}$ and thus $\lim_{\sigma_\mu \rightarrow 0} \lambda^{\text{th}}(x, \bar{\mu}, \sigma_\mu, G) = \frac{-2 \ln \frac{C}{|x|}}{1 - \frac{C}{|x|}}$. As $F(-|x|)$ is decreasing in σ_μ iff $\sigma_\mu \leq \hat{\sigma}(x, \mu, G)$, then $\lambda^{\text{th}}(x, \bar{\mu}, \sigma_\mu, G)$ is increasing in σ_μ up to $\hat{\sigma}(x, \bar{\mu}, G)$ and then decreasing. $\square$

Data availability

No data was used for the research described in the article.

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